Principal Angles and Projection Metrics

Spectral Stability of Equity Covariance

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Abstract

Principal components describe subspaces rather than uniquely determined vectors. Consequently, comparisons of estimated principal directions must be insensitive to changes of orthonormal basis and eigenvector sign. We develop principal angles through the singular value decomposition of the overlap matrix, establish basis invariance, and derive the exact relationship between principal angles and Frobenius distances of orthogonal projectors. A normalized projector distance is then shown to define a metric on the Grassmannian of fixed-dimensional subspaces, providing a geometric measure for studies of principal-subspace stability.

1 Geometric motivation

For two unit vectors $u, v \in \mathbb{R}^d$, the ordinary angle $\alpha \in [0, \pi]$ is determined by $\cos\alpha = u^\mathsf{T} v$. This angle is not an invariant of one-dimensional subspaces: replacing v with $-v$ changes α but does not change the line spanned by v.

Definition 1.1 (Angle between one-dimensional subspaces)**.** For unit vectors $u, v \in \mathbb{R}^d$, the acute angle between $\operatorname{span}(u)$ and $\operatorname{span}(v)$ is

$$\theta = \arccos\big(|u^\mathsf{T} v|\big), \qquad 0 \le \theta \le \frac{\pi}{2}. \tag{1}$$

For example, take

$$u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad v = \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix}, \qquad 0 \le \alpha \le \pi.$$

Then $\theta = \arccos|\cos\alpha|$: the lines coincide at both $\alpha = 0$ and $\alpha = \pi$, while they are perpendicular at $\alpha = \pi/2$.

The corresponding notion for k-dimensional subspaces is a sequence of k *principal angles*. Throughout, assume $1 \le k \le d$ and let

$$U, V \in \mathbb{R}^{d\times k}, \qquad U^\mathsf{T}U = V^\mathsf{T}V = I_k, \qquad \mathcal{U} = \operatorname{im} U, \quad \mathcal{V} = \operatorname{im} V. \tag{2}$$

We write

$$M = U^\mathsf{T} V \in \mathbb{R}^{k\times k} \tag{3}$$

for the *overlap matrix* of two orthonormal bases.

2 Principal angles and the overlap matrix

Lemma 2.1 (Singular values are bounded by one)**.** *Every singular value of $M = U^\mathsf{T}V$ belongs to $[0, 1]$.*

Proof. Let $P_\mathcal{U} = UU^\mathsf{T}$ be the orthogonal projector onto $\mathcal{U}$. Since $P_\mathcal{U}$ is symmetric and idempotent, $0 \preceq P_\mathcal{U} \preceq I_d$. Thus

$$M^\mathsf{T}M = V^\mathsf{T}UU^\mathsf{T}V = V^\mathsf{T}P_\mathcal{U}V \preceq V^\mathsf{T}I_dV = I_k.$$

Moreover $M^\mathsf{T}M \succeq 0$, so its eigenvalues lie in $[0, 1]$. The singular values of M are the nonnegative square roots of these eigenvalues. □

Definition 2.2 (Principal angles)**.** Let $\sigma_1(M) \geq \cdots \geq \sigma_k(M)$ be the singular values of M. The principal angles $0 \leq \theta_1 \leq \cdots \leq \theta_k \leq \pi/2$ between $\mathcal{U}$ and $\mathcal{V}$ are defined by

$$\boxed{\cos\theta_i = \sigma_i(U^\mathsf{T}V), \qquad i = 1, \ldots, k.} \tag{4}$$

The first angle can be understood through an optimization problem over unit vectors in the two subspaces.

Proposition 2.3 (Variational interpretation of the smallest angle)**.**

$$\max_{\substack{x\in\mathcal{U},\, y\in\mathcal{V} \\ \|x\|_2=\|y\|_2=1}} |x^\mathsf{T}y| = \sigma_1(U^\mathsf{T}V) = \cos\theta_1. \tag{5}$$

Proof. Every such pair can be represented as $x = Ua$ and $y = Vb$, with $\|a\|_2 = \|b\|_2 = 1$, because the columns of U and V are orthonormal. Therefore

$$x^\mathsf{T}y = a^\mathsf{T}U^\mathsf{T}Vb = a^\mathsf{T}Mb.$$

For fixed unit b, Cauchy–Schwarz yields $|a^\mathsf{T}Mb| \leq \|Mb\|_2$. Equality holds when $Mb \neq 0$ and $a = Mb/\|Mb\|_2$; if $Mb = 0$, both sides vanish. Consequently,

$$\begin{aligned}\max_{\|a\|_2=\|b\|_2=1} |a^\mathsf{T}Mb| &= \max_{\|b\|_2=1} \|Mb\|_2 \\ &= \|M\|_2 = \sigma_1(M).\end{aligned}$$

Since the cosine is decreasing on $[0, \pi/2]$, maximal alignment corresponds to the smallest acute angle. □

3 Basis invariance and principal vectors

A subspace admits infinitely many orthonormal bases. Its principal angles must therefore be independent of the particular matrices used to represent it.

Theorem 3.1 (Basis invariance)**.** *If $\widetilde{U} = UQ_1$ and $\widetilde{V} = VQ_2$ for orthogonal $Q_1, Q_2 \in \mathbb{R}^{k\times k}$, then*

$$\sigma_i(\widetilde{U}^\mathsf{T}\widetilde{V}) = \sigma_i(U^\mathsf{T}V), \qquad i = 1, \ldots, k.$$

In particular, all principal angles depend only on $\mathcal{U}$ and $\mathcal{V}$.

Proof. The new overlap matrix is

$$\widetilde{U}^\mathsf{T}\widetilde{V} = (UQ_1)^\mathsf{T}(VQ_2) = Q_1^\mathsf{T}MQ_2.$$

Its Gram matrix satisfies

$$\begin{aligned}(\widetilde{U}^\mathsf{T}\widetilde{V})^\mathsf{T}(\widetilde{U}^\mathsf{T}\widetilde{V}) &= Q_2^\mathsf{T}M^\mathsf{T}Q_1Q_1^\mathsf{T}MQ_2 \\ &= Q_2^\mathsf{T}M^\mathsf{T}MQ_2.\end{aligned}$$

Because Q_2 is orthogonal, this matrix is orthogonally similar to $M^\mathsf{T}M$. Their eigenvalues, and hence the singular values of the overlap matrices, coincide. □

The SVD constructs orthonormal pairs realizing the individual principal angles.

Proposition 3.2 (Principal vectors)**.** *Let $M = ASB^\mathsf{T}$ be an SVD with $A, B \in \mathbb{R}^{k\times k}$ orthogonal and $S = \operatorname{diag}(s_1, \ldots, s_k)$, where $s_1 \geq \cdots \geq s_k \geq 0$. Writing a_i, b_i for the columns of A, B, set $x_i = Ua_i \in \mathcal{U}$ and $y_i = Vb_i \in \mathcal{V}$. Then $(x_i)_{i=1}^k$ and $(y_i)_{i=1}^k$ are orthonormal families, and*

$$x_i^\mathsf{T}y_j = s_i\delta_{ij}, \qquad x_i^\mathsf{T}y_i = s_i = \cos\theta_i. \tag{6}$$

Proof. Using $U^\mathsf{T}U = I_k$ and orthogonality of A,

$$x_i^\mathsf{T}x_j = a_i^\mathsf{T}U^\mathsf{T}Ua_j = a_i^\mathsf{T}a_j = \delta_{ij}.$$

The same calculation gives $y_i^\mathsf{T}y_j = \delta_{ij}$. Finally,

$$x_i^\mathsf{T}y_j = a_i^\mathsf{T}U^\mathsf{T}Vb_j = a_i^\mathsf{T}ASB^\mathsf{T}b_j = s_i\delta_{ij}.$$

The diagonal identities follow from Definition 2.2. □

4 Frobenius distance between spectral projectors

Let $P_{\mathcal{U}} = UU^{\mathsf{T}}$ and $P_{\mathcal{V}} = VV^{\mathsf{T}}$. These projectors represent the subspaces without selecting particular directions or signs within them.

Theorem 4.1 (Projector distance identity)**.** *The principal angles between $\mathcal{U}$ and $\mathcal{V}$ satisfy*

$$\boxed{\|P_{\mathcal{U}} - P_{\mathcal{V}}\|_F^2 = 2k - 2\|U^{\mathsf{T}}V\|_F^2 = 2\sum_{i=1}^{k} \sin^2\theta_i.} \tag{7}$$

Proof. For the Frobenius inner product $\langle A, B\rangle_F = \operatorname{tr}(A^{\mathsf{T}}B)$, expanding the squared distance gives

$$\|P_{\mathcal{U}} - P_{\mathcal{V}}\|_F^2 = \|P_{\mathcal{U}}\|_F^2 + \|P_{\mathcal{V}}\|_F^2 - 2\langle P_{\mathcal{U}}, P_{\mathcal{V}}\rangle_F.$$

The projectors are symmetric and idempotent. Therefore,

$$\|P_{\mathcal{U}}\|_F^2 = \operatorname{tr}(P_{\mathcal{U}}^2) = \operatorname{tr}(P_{\mathcal{U}}) = \operatorname{tr}(U^{\mathsf{T}}U) = k,$$

and likewise $\|P_{\mathcal{V}}\|_F^2 = k$. By cyclicity of the trace,

$$\langle P_{\mathcal{U}}, P_{\mathcal{V}}\rangle_F = \operatorname{tr}(UU^{\mathsf{T}}VV^{\mathsf{T}}) = \operatorname{tr}(V^{\mathsf{T}}UU^{\mathsf{T}}V) = \|U^{\mathsf{T}}V\|_F^2.$$

Hence $\|P_{\mathcal{U}} - P_{\mathcal{V}}\|_F^2 = 2k - 2\|M\|_F^2$. Now let $M = ASB^{\mathsf{T}}$; then

$$\begin{aligned}\|M\|_F^2 &= \operatorname{tr}(M^{\mathsf{T}}M) = \operatorname{tr}(BS^{\mathsf{T}}SB^{\mathsf{T}}) = \operatorname{tr}(S^{\mathsf{T}}S)\\ &= \sum_{i=1}^{k} s_i^2 = \sum_{i=1}^{k} \cos^2\theta_i.\end{aligned}$$

Substituting and using $1 - \cos^2\theta_i = \sin^2\theta_i$ proves the result. □

Example 4.2 (Different angle profiles)**.** Consider two pairs of five-dimensional subspaces with principal angles, in degrees,

	θ_1	θ_2	θ_3	θ_4	θ_5
Pair A	2	3	4	5	6
Pair B	2	3	4	5	70

Such profiles can be realized, for example, in sufficiently large ambient dimension. Equation (7) shows that a single poorly aligned direction can increase the total projector distance considerably, even when the remaining directions are closely aligned.

5 A normalized distance on the Grassmannian

The dimension-dependent scale of $\|P_{\mathcal{U}} - P_{\mathcal{V}}\|_F$ motivates a normalized measure. Denote by $\operatorname{Gr}(k, d)$ the set of all k-dimensional linear subspaces of $\mathbb{R}^d$.

Definition 5.1 (Normalized projection distance)**.** For $\mathcal{U}, \mathcal{V} \in \operatorname{Gr}(k, d)$, define

$$\boxed{d_k(\mathcal{U}, \mathcal{V}) = \frac{\|P_{\mathcal{U}} - P_{\mathcal{V}}\|_F}{\sqrt{2k}} = \left(\frac{1}{k}\sum_{i=1}^{k} \sin^2\theta_i\right)^{1/2} = \left(1 - \frac{1}{k}\sum_{i=1}^{k} \sigma_i(U^{\mathsf{T}}V)^2\right)^{1/2}.} \tag{8}$$

Proposition 5.2 (Range and equality cases)**.** *For every $\mathcal{U}, \mathcal{V} \in \operatorname{Gr}(k, d)$,*

$$0 \le d_k(\mathcal{U}, \mathcal{V}) \le 1. \tag{9}$$

Moreover $d_k(\mathcal{U}, \mathcal{V}) = 0$ precisely when $\mathcal{U} = \mathcal{V}$, while $d_k(\mathcal{U}, \mathcal{V}) = 1$ precisely when $\mathcal{U} \perp \mathcal{V}$. The latter case is attainable only if $d \ge 2k$.

Proof. By Lemma 2.1, every squared singular value of M belongs to $[0,1]$. Thus $0 \le \sum_i \sigma_i(M)^2 \le k$, and the range follows from (8). If $d_k = 0$ then $P_{\mathcal{U}} = P_{\mathcal{V}}$, so equality of their images gives $\mathcal{U} = \mathcal{V}$. Conversely, if the subspaces agree, then their orthogonal projectors agree and $d_k = 0$.

For completeness, when $\mathcal{U} = \mathcal{V}$ the basis matrices satisfy $V = UQ$ for some $Q \in \mathbb{R}^{k\times k}$. As $I_k = V^\mathsf{T}V = Q^\mathsf{T}Q$, the matrix Q is orthogonal, and hence $VV^\mathsf{T} = UQQ^\mathsf{T}U^\mathsf{T} = UU^\mathsf{T}$.

Finally, $d_k = 1$ if and only if every singular value of $U^\mathsf{T}V$ vanishes, equivalently $U^\mathsf{T}V = 0$. This means all vectors in $\mathcal{U}$ are orthogonal to all vectors in $\mathcal{V}$. Two orthogonal k-dimensional subspaces require an ambient dimension of at least $2k$. □

Theorem 5.3 (Metric property)**.** *For fixed $k \ge 1$, the function d_k is a metric on* $\mathrm{Gr}(k,d)$.

Proof. Nonnegativity and definiteness follow from Proposition 5.2. Symmetry follows from

$$\|P_{\mathcal{U}} - P_{\mathcal{V}}\|_F = \|P_{\mathcal{V}} - P_{\mathcal{U}}\|_F.$$

For any $\mathcal{U}, \mathcal{V}, \mathcal{W} \in \mathrm{Gr}(k,d)$, the Frobenius-norm triangle inequality gives

$$\|P_{\mathcal{U}} - P_{\mathcal{W}}\|_F \le \|P_{\mathcal{U}} - P_{\mathcal{V}}\|_F + \|P_{\mathcal{V}} - P_{\mathcal{W}}\|_F.$$

Dividing by $\sqrt{2k}$ yields $d_k(\mathcal{U},\mathcal{W}) \le d_k(\mathcal{U},\mathcal{V}) + d_k(\mathcal{V},\mathcal{W})$. □

Remark 5.4 (Interpretation across dimensions). Normalization bounds d_k by one for every k, but it does not by itself make estimates for different subspace dimensions statistically interchangeable. Sampling variability, the relevant eigengap, and dimensional constraints can change with k. Comparisons across k therefore require appropriate experimental controls.

6 Connection to covariance spectral stability

Suppose $\widehat{U}_t \in \mathbb{R}^{d\times k}$ contains orthonormal leading eigenvectors of a covariance estimate in a rolling window indexed by t. The associated principal subspace is $\widehat{\mathcal{U}}_t = \operatorname{im} \widehat{U}_t$. A natural descriptive measure of adjacent-window subspace drift is

$$D_{t,k} = d_k(\widehat{\mathcal{U}}_t, \widehat{\mathcal{U}}_{t+1}) = \frac{\|\widehat{U}_t\widehat{U}_t^\mathsf{T} - \widehat{U}_{t+1}\widehat{U}_{t+1}^\mathsf{T}\|_F}{\sqrt{2k}}. \tag{10}$$

The metric is invariant to orthonormal rotations within each estimated subspace, including individual eigenvector sign changes. A change in $D_{t,k}$ nevertheless need not imply an economic regime shift: it may arise from sampling noise, changes in volatility scaling, estimation choices, or genuine changes in the population covariance structure. Distinguishing these mechanisms requires separate experimental analysis.